ALGEBRA 1 · UNIT 12: STATISTICS AND DATA MODELING

Screen Time Study

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students act as researchers studying teen screen time and sleep. They compare two classes with dot plots, box plots, and measures of center and spread, analyze a two-way table, fit a line to scatter-plot data, interpret correlation, and design a survey study of their own.

One theme connects the whole statistics unit: distributions, two-way tables, and bivariate data. The final conclusion requires students to separate association from causation, a key modeling skill.

Standards Addressed

- S-ID.1–3 Represent and compare distributions (center, spread, outliers)
- S-ID.5 Two-way frequency tables and relative frequencies
- S-ID.6 Scatter plots, lines of fit, and residuals
- S-ID.7–9 Interpret slope, intercept, correlation, and causation

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch: How many hours of sleep did you get last night? Make a quick class dot plot. Part 1 compares the two classes.
Day 2	Part 2: two-way table and scatter plot with a line of fit.
Day 3	Part 3: conclusions and the research poster for a new study.

Materials

- Student handout
- Graphing calculators or Desmos (for Honors regression)
- Rulers
- Optional: an anonymous class survey

How the Three Versions Differ

Level	What changes
CP2	Dot plots, five-number summary, mean and median, totals and simple percents in a two-way table, and predictions from a given line of fit.
CP1	IQR and outliers with the 1.5 × IQR rule, MAD, conditional relative frequencies, a line of fit from two points, and correlation vs. causation.
Honors	Standard deviation, the effect of outliers, regression and residuals with technology, the correlation coefficient, and a study design with a lurking-variable plan.

Watch For (Common Misconceptions)

- Saying the class with the higher maximum has 'more screen time' overall.
- Using row totals and column totals interchangeably for conditional percents.
- Concluding that screen time causes less sleep from correlational data.

Extension Ideas

- Collect anonymous class data and repeat the analysis.
- Compare weekday and weekend data.
- Research how real health studies are designed.

Answer Key – CP2

Part 1: Compare the Classes

#	Task	Answer
1	Make a dot plot of Class A's data.	Dots stacked above each value: one at 2, two at 3, three at 4, three at 5, two at 6, two at 7, one at 8, one at 12.
2	Find the mean and the median of each class.	Class A: mean 81 ÷ 15 = 5.4, median 5. Class B: mean 60 ÷ 15 = 4, median 4.
3	Find the five-number summary (minimum, Q1, median, Q3, maximum) of each class and draw both box plots on the same scale.	A: 2, 4, 5, 7, 12. B: 1, 3, 4, 5, 9.
4	Which class has more screen time overall? Use the medians and the box plots to explain.	Class A. Its median (5) is higher than Class B's (4), and its whole box is shifted to the right.

Part 2: Tables & Trends

#	Task	Answer
5	Complete the totals in the table.	Row totals 70 and 50; column totals 52 and 68; grand total 120.
6	What percent of all 120 students sleep less than 8 hours?	68 ÷ 120 ≈ 56.7%
7	Of the students with 4 or more hours of screen time, what percent sleep less than 8 hours? Of those with less than 4 hours?	52 ÷ 70 ≈ 74.3% of heavy screen users; 16 ÷ 50 = 32% of light users.
8	Describe the association in the scatter plot (direction and strength). Then use the line of fit $y = -0.4x + 9.8$ to predict the sleep of a student…	Strong negative linear association. $y = -0.4(6) + 9.8 = 7.4$ hours

Part 3: Your Own Study

#	Task	Answer
9	Write a statistical question about two variables for students at your school.	Answers vary.
10	Describe how you would collect the data and which display you would use. Explain why.	Answers vary.
11	Does the screen-time data prove that screens cause less sleep? Give another reason students who use screens more might sleep less.	No. Other factors such as homework, sports, jobs, or stress could affect both.

Answer Key – CP1

Part 1: Compare the Classes

#	Task	Answer
1	Find the mean, median, IQR, and range for each class.	A: mean 5.4, median 5, IQR 7 − 4 = 3, range 10. B: mean 4, median 4, IQR 5 − 3 = 2, range 8.
2	Use the 1.5 × IQR rule to find any outliers in each class.	A: fences 4 − 4.5 = −0.5 and 7 + 4.5 = 11.5 → 12 is an outlier. B: fences 0 and 8 → 9 is an outlier.
3	Find the mean absolute deviation (MAD) of Class B. Which class is more spread out?	MAD of B = 22 ÷ 15 ≈ 1.47 hours. Class A (MAD ≈ 1.81, IQR 3) is more spread out.
4	Draw both box plots on one scale and write a comparison of shape, center, and spread.	Both are slightly skewed right with a high outlier. A has a higher center (5 vs. 4) and more spread (IQR 3 vs. 2).

Part 2: Tables & Trends

#	Task	Answer
5	Complete the table. Find the conditional relative frequency of sleeping less than 8 hours for each screen-time group. Is there an association?...	Heavy users: 52 ÷ 70 ≈ 74.3%. Light users: 16 ÷ 50 = 32%. Yes, the percents are very different, so there is an association.
6	Draw a line of fit through (2, 9.0) and (10, 5.8). Write its equation.	Slope (5.8 − 9.0) ÷ (10 − 2) = −0.4; y = −0.4x + 9.8
7	Interpret the slope and y-intercept of your line in context.	Slope: each extra hour of screen time goes with about 0.4 hour (24 minutes) less sleep. Intercept: a student with no screen time is predicted to sleep 9.8 hours.
8	Estimate the correlation coefficient r (choose: −0.99, −0.5, 0, 0.5, 0.99) and explain.	r ≈ −0.99: the points fall very close to a line that goes down.

Part 3: Conclusions & Your Study

#	Task	Answer
9	A news headline says: "Screens Steal Sleep!" Explain whether this data supports that claim.	No. The data show an association, not cause and effect. This was an observational survey, and lurking variables may explain both.
10	Use your line to predict sleep for 12 hours of screen time. Why is this prediction less reliable than one for 6 hours?	y = −0.4(12) + 9.8 = 5 hours. 12 is outside the data (extrapolation), so the pattern may not continue.
11	Design your own two-variable study: question, sample, data collection, display, and the statistic you'll report.	Answers vary.

Answer Key – Honors

Part 1: Compare the Classes

#	Task	Answer
1	Find the mean, median, IQR, and standard deviation of each class (state whether you use σ or s).	A: mean 5.4, median 5, IQR 3, σ ≈ 2.39 (s ≈ 2.47). B: mean 4, median 4, IQR 2, σ ≈ 1.93 (s = 2.00).
2	Identify outliers with the 1.5 × IQR rule. Remove Class A's outlier and recompute the mean and median. What changed and why?	Outliers: 12 in A, 9 in B. Without 12: mean ≈ 4.93, median still 5. The mean dropped; the median is resistant to outliers.
3	Which measures of center and spread are most appropriate for these data? Justify.	Median and IQR, because both distributions are skewed right with outliers.
4	Write a comparison of the two classes that a non-statistician would understand.	Answers vary: Class A students typically use screens about 1 hour more per day and vary more.

Part 2: Model the Relationship

#	Task	Answer
5	From the two-way table, find the joint, marginal, and conditional relative frequencies needed to compare sleep for the two screen-time groups. Is...	Joint: 18, 52, 34, 16 out of 120 (15%, 43.3%, 28.3%, 13.3%). Marginal: 58.3% heavy users, 41.7% light. Conditional: 74.3% vs. 32% sleep less than 8 h. Yes.
6	Use technology to find the least-squares regression line and r. Round to the nearest hundredth.	$\hat{y} \approx -0.42x + 9.82$, $r \approx -0.99$
7	Find the residual for the student with 7 hours of screen time and 6.6 hours of sleep. Interpret it.	Predicted $\hat{y} \approx -0.42(7) + 9.82 \approx 6.87$ (using unrounded values, 6.87). Residual ≈ 6.6 − 6.87 ≈ −0.27: the student sleeps about 16 minutes less than predicted.
8	Describe what a residual plot for these data would look like if a linear model is appropriate.	Points scattered randomly above and below 0 with no curve or pattern.

Part 3: Design a Better Study

#	Task	Answer
9	Explain why the survey cannot establish causation. Name two lurking variables.	It was observational (no random assignment). Lurking variables: homework load, sports schedules, jobs, stress, caffeine.
10	Design an experiment (with random assignment) that could test whether screen time before bed affects sleep. Address ethics and practicality.	Answers vary: randomly assign volunteers to no screens or screens for 1 hour before bed for two weeks; measure sleep with a tracker; get consent.

11	Write the conclusion section of your research poster, including the model, r, a residual, and limits of the study.	Answers vary.

Scoring Rubric (all levels)

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Original study has a clear question, a sensible sampling plan, and appropriate displays and statistics. Goes beyond the minimum with a creative, realistic choice.	Original study has a clear question, a sensible sampling plan, and appropriate displays and statistics.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Research poster shows a question, displays with summaries, a model with interpretation, and a conclusion that respects correlation vs. causation. Polished, organized, and visually engaging.	Research poster shows a question, displays with summaries, a model with interpretation, and a conclusion that respects correlation vs. causation. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.